

DiffMI Evaluation on FaceNet

Objective:

The goal of this experiment was to evaluate the effectiveness of the **DiffMI (Diffusion-based Membership Inference)** attack against a FaceNet face recognition model and analyze whether the generated reconstructions generalize across models.

Experimental Setup:

- Target Model: FaceNet
- Attack Method: DiffMI (White-box setting)
- Dataset: FaceScrub identities (<https://www.kaggle.com/datasets/rajnishe/facescrub-full>) (Paper: <https://ieeexplore.ieee.org/abstract/document/7025068>)
- Latent optimization performed using DiffMI inversion pipeline
- Generated reconstructions were evaluated on:
 - FaceNet (same model used during attack)
 - ArcFace (cross-model evaluation)

Results

1. FaceNet Verification Performance

Metric	Value
EER (Equal Error Rate)	0.0507
Verification Threshold	0.3674

Interpretation

- The FaceNet model achieves an EER of approximately **5.07%**.
- The threshold used for identity verification is **0.3674**.
- This provides the decision boundary used during attack evaluation.

2. DiffMI Attack Performance

Metric	Value
Attack Success Rate (ASR)	96.0%
Type I Accuracy	100%
Type II Accuracy	95.0%

Interpretation

- The attack successfully reconstructed target identities in **96% of cases**.
- Perfect Type I accuracy indicates that genuine target matches were consistently recovered.
- High Type II accuracy suggests strong separation between target and non-target identities.
- Overall, DiffMI demonstrates very high effectiveness against FaceNet in the tested setting.

3. Reconstruction Quality

Metric	Value
Similarity Score	0.9832

Interpretation

- The reconstructed images are extremely close to the target identity embeddings.
- A similarity score above **0.98** indicates that DiffMI can recover highly representative facial features of the target individual.

4. Computational Cost

Metric	Value
Total Attack Time	13,155.84 seconds
Approximate Runtime	3.65 hours

Interpretation

- The attack is computationally expensive.
- Although highly effective, reconstruction requires several hours of optimization and diffusion sampling.

Cross-Model Evaluation (FaceNet -> ArcFace)

To evaluate whether reconstructed identities generalize beyond the attacked model, the recovered images were tested using ArcFace embeddings.

Results

Metric	Value
ArcFace Acc@1	74%
ArcFace Acc@5	90%
Feature Distance	22.61

Interpretation

- Even though the attack targeted FaceNet, the reconstructed identities remain recognizable by ArcFace.
- An **Acc@1 of 74%** means the correct identity is the top prediction in nearly three-quarters of cases.
- An **Acc@5 of 90%** indicates that the correct identity appears among the top five predictions most of the time.
- This suggests that DiffMI is recovering identity-related information rather than merely overfitting to FaceNet-specific embeddings.
- The reconstructions exhibit a meaningful degree of cross-model transferability.

Key Findings

1. DiffMI achieves a very high attack success rate (**96% ASR**) against FaceNet.
2. Reconstructed identities closely match target embeddings (**0.983 similarity**).
3. FaceNet is vulnerable to identity reconstruction under the evaluated white-box threat model.
4. The recovered identities generalize reasonably well to ArcFace (**74% Acc@1, 90% Acc@5**).
5. The attack is effective but computationally expensive (~3.65 hours for the experiment).
6. Preliminary evidence suggests that DiffMI recovers genuine identity information rather than exploiting model-specific artifacts.

Current Status

- FaceNet attack completed and evaluated.
- Cross-model evaluation using ArcFace completed.
- Full ArcFace attack experiment is currently being rerun because the execution environment terminated after approximately 41 minutes of inactivity.
- Next will perform the same DiffMI attack pipeline directly on ArcFace and compare attack effectiveness, reconstruction quality, and cross-model transferability between FaceNet and ArcFace.